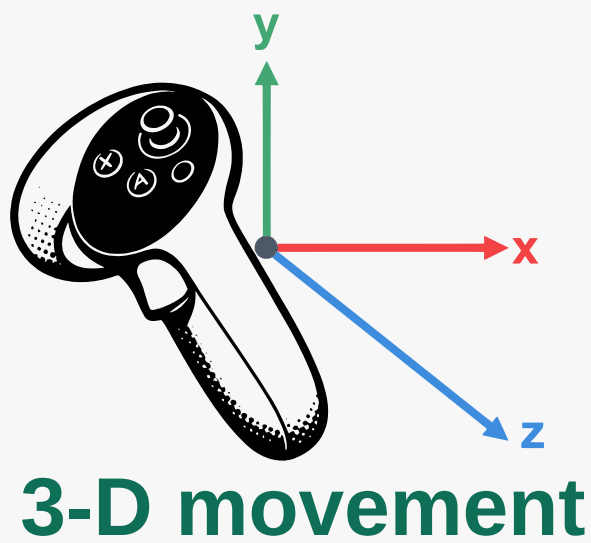


**1**

Capture 3-D belly movement

Abdomen-mounted controller pose read each frame.

**2**

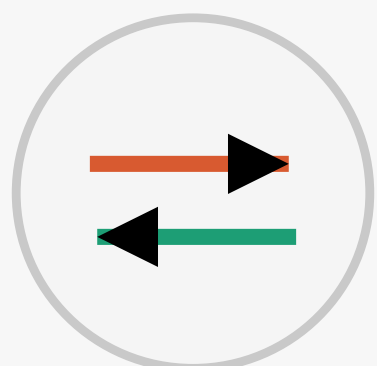
Filter to 1-D breathing signal

Projected onto calibrated axis. Range filter rejects non-respiratory motion. Dual smoothing yields 1-D trace.

**3**

Detect breath phase

Positive slope = **inhale**, negative = exhale, near-zero = hold. Signal reduced to three categorical states.

**4**

Map phase to radius dynamic

Inhale drives expansion, exhale drives contraction, hold maintains radius.

**5**

Integrate at fixed rate

Velocity set so full traversal takes the guided phase duration (here, 4 s).